

COMMANDS 2 – Closing the Section 7.2 Limitations

(each block maps to one limitation sentence; run blocks in order)

All commands for ****Anaconda Prompt (Windows)****, run from the repo root
`D:\Courses\hdpba-hfl`. Sweeps resume on re-run (finished runs skip).
Single `train` invocations do not resume, so let each finish.

Run-count summary: B 21 · C 80 · D 30 · E 0 (reuses D) · F 30 · G 12,
total 173 training runs plus two script-only blocks (H, I).
Multiply by your Block-A pilot minutes for the schedule. Suggested split:
machine 1 takes B, D, F; machine 2 takes C, G. Never split one block
across machines (keeps the data-blind canonical rule trivial).

Discipline (Section 6.2 of the manuscript): the full set below is the
announced cell list. Every cell is reported, whichever way it comes out.
No selective re-running after peeking at outcomes.

A. Environment sanity and timing pilot (once per machine, ~25 min)

```
```bat
conda activate hdpba
cd D:\Courses\hdpba-hfl
python -m pytest tests\ -q
:: expect: 32 passed

:: timing pilot; this run also counts towards Block B (P1.e, seed 2024)
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method
escs --seed 2024 --outdir runs\r19_mnist_seedfill\assignment-random_method-
escs\seed2024 --name assignment-random_method-escs_s2024
:: note wall-clock minutes; send me summary.json + the console log from
:: this run so the ledger ingestion script can be built while B-C run
```
```

Verify the federation-size flags for Block G before scheduling it:

```
```bat
python -m hdpba_hfl.train --help
:: confirm the flag names assumed in Block G (--num_clients --num_edges)
```
```

B. Limitation: "H-DPBA and oracle arms are single-seed at the task ## level" – MNIST seed completion (21 runs incl. the Block-A pilot)

Closes the single-seed rows of Table 6 and enables the pre-registered Wilcoxon and non-inferiority tests (H-DPBA vs oracle, margin one point).

```
```bat
```

```
:: B.1 H-DPBA + ESCS, remaining seeds
```

```
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method escs
--seed 123 --outdir runs\r19_mnist_seedfill\assignment-hdpba_method-
escs\seed123 --name assignment-hdpba_method-escs_s123
```

```
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method escs
--seed 2024 --outdir runs\r19_mnist_seedfill\assignment-hdpba_method-
escs\seed2024 --name assignment-hdpba_method-escs_s2024
```

```
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method escs
--seed 7 --outdir runs\r19_mnist_seedfill\assignment-hdpba_method-escs\seed7 -
-name assignment-hdpba_method-escs_s7
```

```
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method escs
--seed 99 --outdir runs\r19_mnist_seedfill\assignment-hdpba_method-escs\seed99
--name assignment-hdpba_method-escs_s99
```

```
:: B.2 Oracle + ESCS, remaining seeds
```

```
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment oracle --method
escs --seed 123 --outdir runs\r19_mnist_seedfill\assignment-oracle_method-
escs\seed123 --name assignment-oracle_method-escs_s123
```

```
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment oracle --method
escs --seed 2024 --outdir runs\r19_mnist_seedfill\assignment-oracle_method-
escs\seed2024 --name assignment-oracle_method-escs_s2024
```

```
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment oracle --method
escs --seed 7 --outdir runs\r19_mnist_seedfill\assignment-oracle_method-
escs\seed7 --name assignment-oracle_method-escs_s7
```

```
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment oracle --method
escs --seed 99 --outdir runs\r19_mnist_seedfill\assignment-oracle_method-
escs\seed99 --name assignment-oracle_method-escs_s99
```

```
:: B.3 AUTO + random, five seeds (selector under random placement)
```

```
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method
auto --seed 42 --outdir runs\r19_mnist_seedfill\assignment-random_method-
auto\seed42 --name assignment-random_method-auto_s42
```

```
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method
```

```

auto --seed 123 --outdir runs\r19_mnist_seedfill\assignment-random_method-
auto\seed123 --name assignment-random_method-auto_s123
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method
auto --seed 2024 --outdir runs\r19_mnist_seedfill\assignment-random_method-
auto\seed2024 --name assignment-random_method-auto_s2024
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method
auto --seed 7 --outdir runs\r19_mnist_seedfill\assignment-random_method-
auto\seed7 --name assignment-random_method-auto_s7
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method
auto --seed 99 --outdir runs\r19_mnist_seedfill\assignment-random_method-
auto\seed99 --name assignment-random_method-auto_s99

```

:: B.4 H-DPBA + AUTO, five seeds (the full proposed system on MNIST)

```

python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method auto
--seed 42 --outdir runs\r19_mnist_seedfill\assignment-hdpba_method-auto\seed42
--name assignment-hdpba_method-auto_s42
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method auto
--seed 123 --outdir runs\r19_mnist_seedfill\assignment-hdpba_method-
auto\seed123 --name assignment-hdpba_method-auto_s123
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method auto
--seed 2024 --outdir runs\r19_mnist_seedfill\assignment-hdpba_method-
auto\seed2024 --name assignment-hdpba_method-auto_s2024
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method auto
--seed 7 --outdir runs\r19_mnist_seedfill\assignment-hdpba_method-auto\seed7 -
--name assignment-hdpba_method-auto_s7
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method auto
--seed 99 --outdir runs\r19_mnist_seedfill\assignment-hdpba_method-auto\seed99
--name assignment-hdpba_method-auto_s99

```

:: B.5 ESCS + random, remaining seeds (pilot covered 2024)

```

python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method
escs --seed 7 --outdir runs\r19_mnist_seedfill\assignment-random_method-
escs\seed7 --name assignment-random_method-escs_s7
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method
escs --seed 99 --outdir runs\r19_mnist_seedfill\assignment-random_method-
escs\seed99 --name assignment-random_method-escs_s99
...

```

Note: random+ESCA at edge\_frac 1.0 arrives from Block C (five seeds), so it is deliberately absent here. If Block C's base config turns out to differ from e1\_main in any training parameter (check below), add the five random+esca runs here in the same pattern.

Quicklook after the block:

```
```bat
python -m hdpba_hfl.stats runs\r19_mnist_seedfill --metric final_acc
python -m hdpba_hfl.stats runs\r19_mnist_seedfill --metric final_acc --compare
assignment-hdpba assignment-oracle
```
```

---

**## C. Limitation: "the dropout-robustness prediction of Remark 2 ...  
## not yet evaluated" – edge-participation sweep (80 runs)**

Uses your existing pre-registered sweep unchanged. Before launching, confirm ``configs/e6_dropout.yaml`` matches the Table 4 defaults of ``configs/e1_main.yaml`` (T, U, E, K, J, eps\_a, sigma); if anything differs, stop and report the difference before running.

```
```bat
python scripts\sweep.py sweeps\e6_edge_dropout.yaml --outdir runs\r20_dropout
python -m hdpba_hfl.stats runs\r20_dropout --metric final_acc
```
```

Pre-registered verdicts (already frozen in analyze\_new\_runs.py):  
primary contrast slope(A4)/slope(A0) under ESCS  $\leq 0.6$ , secondary the same under ESCA. The script prints PASS or FAIL per contrast.

---

**## D. Limitation: "one dataset at a saturated operating point" –  
## Fashion-MNIST full grid (30 runs)**

One product sweep covers the main comparison AND Block E's factorial. Create ``sweeps\r21_fmnist_full.yaml`` with exactly this content (same schema as your e6 sweep file):

```
```yaml
# r21 -- FMNIST main comparison + assignment-by-weighting factorial.
base: configs/e1_main.yaml
seeds: [42, 123, 2024]
grid:
  dataset: [fmnist]
  assignment: [random, hdpba]
```

```
method: [bl1, bl2, escs, esca, auto]
```
```

Then:

```
```bat
python scripts\sweep.py sweeps\r21_fmnist_full.yaml --outdir
runs\r21_fmnist_full
python -m hdpba_hfl.stats runs\r21_fmnist_full --metric final_acc
```
```

(Equivalent alternative per your COMMANDS.md section 4 workflow: edit the ``fixed:`` block of ``sweeps\e1_main.yaml`` to fmnist and sweep to the same outdir. Use one route only.)

---

```
E. Limitation: "the negative interaction between assignment and
accuracy weighting implied by Remark 3 ... not yet evaluated"
(0 new runs)
```

The  $2 \times 3$  factorial (assignment {random, hdpba} x central rule {uniform = bl2, escs, esca} at fixed edge-tier sample-size weighting) is a subset of Block D's grid. After ingesting Block D:

```
```bat
python -m hdpba_hfl.stats runs\r21_fmnist_full --metric final_acc --compare
method-esca method-escs
python analyze_new_runs.py --data path\to\ledger_csvs --out analysis_out
:: analyze_new_runs.py computes the pre-registered interaction contrast
:: (ESCA-ESCS gap under A4 minus under A0; predicted negative) and
:: writes tables\tab_ablation.tex
```
```

---

```
F. Multi-dataset confirmation on CIFAR-10 (30 runs; the slow block,
run last)
```

Create ``sweeps\r24_cifar_full.yaml``:

```
```yaml
# r24 -- CIFAR-10 main comparison + factorial (mirror of r21).
base: configs/e1_main.yaml
seeds: [42, 123, 2024]
grid:
  dataset: [cifar10]
  assignment: [random, hdpba]
```

```

    method: [b11, b12, escs, esca, auto]
'''
'''bat
python scripts\sweep.py sweeps\r24_cifar_full.yaml --outdir
runs\r24_cifar_full
python -m hdpba_hfl.stats runs\r24_cifar_full --metric final_acc
'''

---

## G. Limitation: "scalability beyond K = 100" (12 runs)
## [VERIFY the two flag names via Block A's --help before running]

Capacity kappa = 5 is preserved by scaling J = K/5. Flag names below are
assumed; correct them to whatever `--help` shows.

'''bat
:: K = 200, J = 40
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method
auto --seed 42 --num_clients 200 --num_edges 40 --outdir
runs\r23_scale\K200_assignment-random\seed42 --name K200_random_s42
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method
auto --seed 123 --num_clients 200 --num_edges 40 --outdir
runs\r23_scale\K200_assignment-random\seed123 --name K200_random_s123
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method
auto --seed 2024 --num_clients 200 --num_edges 40 --outdir
runs\r23_scale\K200_assignment-random\seed2024 --name K200_random_s2024
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method auto
--seed 42 --num_clients 200 --num_edges 40 --outdir
runs\r23_scale\K200_assignment-hdpba\seed42 --name K200_hdpba_s42
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method auto
--seed 123 --num_clients 200 --num_edges 40 --outdir
runs\r23_scale\K200_assignment-hdpba\seed123 --name K200_hdpba_s123
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method auto
--seed 2024 --num_clients 200 --num_edges 40 --outdir
runs\r23_scale\K200_assignment-hdpba\seed2024 --name K200_hdpba_s2024

:: K = 500, J = 100
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method

```

```

auto --seed 42 --num_clients 500 --num_edges 100 --outdir
runs\r23_scale\K500_assignment-random\seed42 --name K500_random_s42
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method
auto --seed 123 --num_clients 500 --num_edges 100 --outdir
runs\r23_scale\K500_assignment-random\seed123 --name K500_random_s123
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment random --method
auto --seed 2024 --num_clients 500 --num_edges 100 --outdir
runs\r23_scale\K500_assignment-random\seed2024 --name K500_random_s2024
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method auto
--seed 42 --num_clients 500 --num_edges 100 --outdir
runs\r23_scale\K500_assignment-hdpba\seed42 --name K500_hdpba_s42
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method auto
--seed 123 --num_clients 500 --num_edges 100 --outdir
runs\r23_scale\K500_assignment-hdpba\seed123 --name K500_hdpba_s123
python -m hdpba_hfl.train --config configs/e1_main.yaml --dataset mnist --
partition pathological --classes_per_client 2 --assignment hdpba --method auto
--seed 2024 --num_clients 500 --num_edges 100 --outdir
runs\r23_scale\K500_assignment-hdpba\seed2024 --name K500_hdpba_s2024
```

```

---

**## H. Limitation: "attack-based privacy evaluation on both channels"**  
**## (script work, no training runs; second wave)**

Assignment channel: a transcript attack is post-processing of the already-logged assignment transcripts (noised aggregates + choices).  
Model channel: a loss-threshold membership inference first pass reuses the trained checkpoints from Blocks B, D, F. Neither harness exists in the repo yet. Send me one run directory listing (files inside one seed folder, e.g. ``dir runs\r19_mnist_seedfill\assignment-hdpba_method-escs\seed42``)

and I will write both harnesses against what the runs actually save.  
LiRA-grade MIA (shadow models) is deliberately deferred; it costs more compute than Blocks B-G combined.

---

**## I. After every block: ingest, analyse, hand off**

```

```bat
:: 1. append the block's rows to runs_master.csv / rounds_long.csv
:: (ingest_runs.py -- I build this from your Block-A pilot's
:: summary.json + console log; until then, send raw logs)

```

```
:: 2. regenerate all tables, figures, and pre-registered verdicts
python analyze_new_runs.py --data path\to\ledger_csvs --out analysis_out

:: 3. read analysis_out\stats_report.txt (PASS / FAIL / PENDING per test)
:: 4. send the updated CSVs (or stats_report.txt + tables\ ) for the
::    manuscript integration
```
```

---

## ## Mapping back to the manuscript

- Block B closes: Table 6 preliminary rows, Wilcoxon + non-inferiority.
- Block C closes: Remark 2 / Section 6.6 (new figure fig\_dropout).
- Blocks D + F close: "one dataset", "baselines saturate"; new Tables.
- Block E closes: Remark 3 interaction (tab\_ablation).
- Block G closes: "scalability beyond  $K = 100$ ".
- Block H closes: "attack-based privacy evaluation" (second wave).

Each closure moves one sentence out of Section 7.2 and its numbers into Section 6; the prose follows the pre-registered verdicts either way.